iLearn Practice Exam

12BR · Chapter / Topic 16

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Topic 16: System Software

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 72

Marks obtained: ..... / 72

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 72.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 11 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 Describe the process of executing a program using an interpreter.

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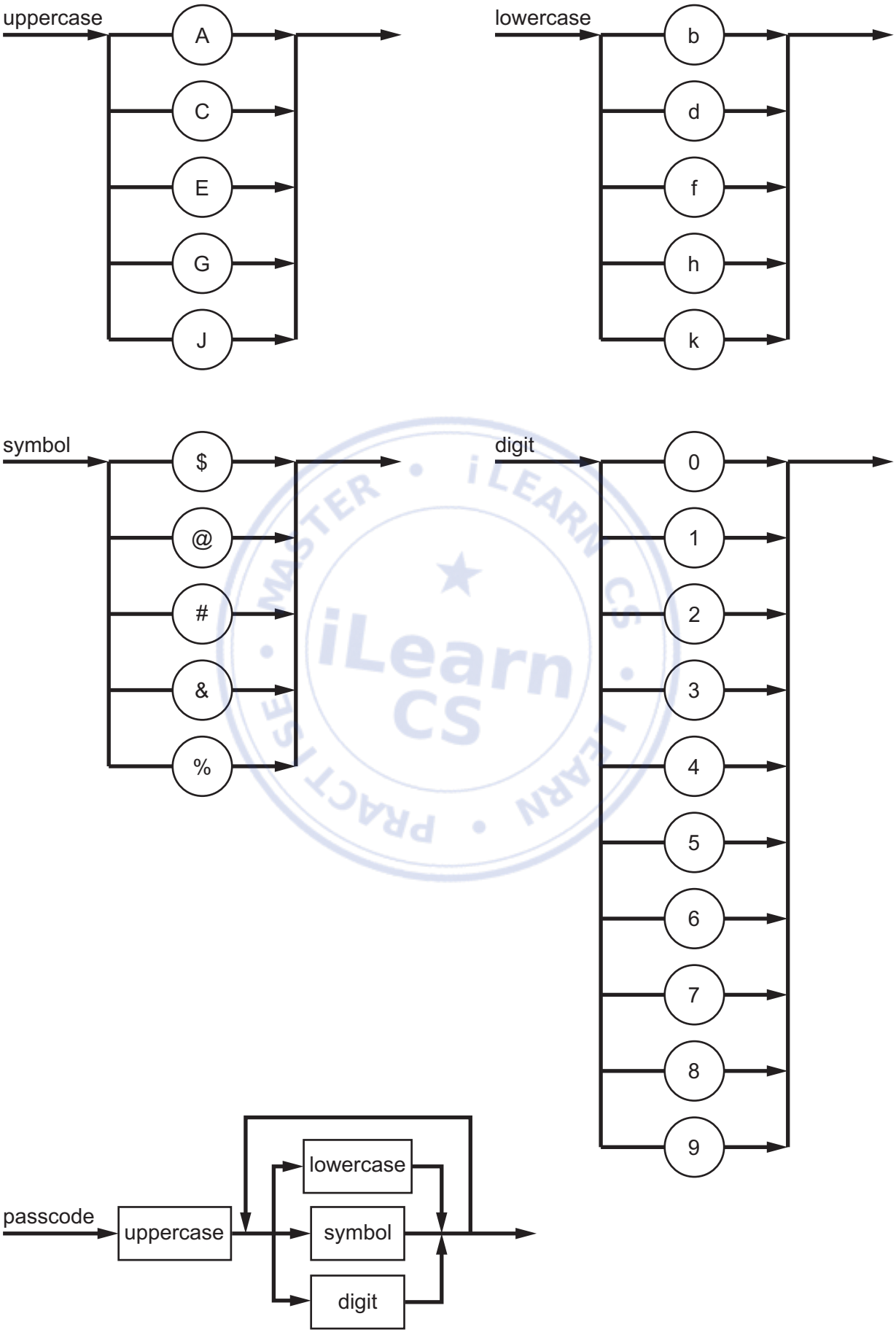
.....

.....

..... [4]

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2 Several syntax diagrams are shown.



(a) State why each **passcode** is invalid for the given syntax diagrams.

`#Jd7`

Reason .....

.....

`C%6A`

Reason .....

.....

[2]

(b) Complete the Backus-Naur Form (BNF) for `<uppercase>` and `<passcode>`.

`<uppercase> ::=` .....

.....

`<passcode> ::=` .....

.....

.....

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[4]

3 **(a)** Describe what is meant by **multi-tasking** and how it benefits process management.

.....

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.....

..... [2]

(b) Explain the function of the shortest remaining time scheduling routine **and** give a benefit of this routine.

Function .....

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Benefit .....

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[4]

4 **(a)** Outline the process of using a stack to evaluate an expression written using Reverse Polish Notation (RPN).

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..... [3]

(b) Write the following infix expression in RPN.

```
((18 - 15 + 3) * 5) / 12
```

.....

.....

.....

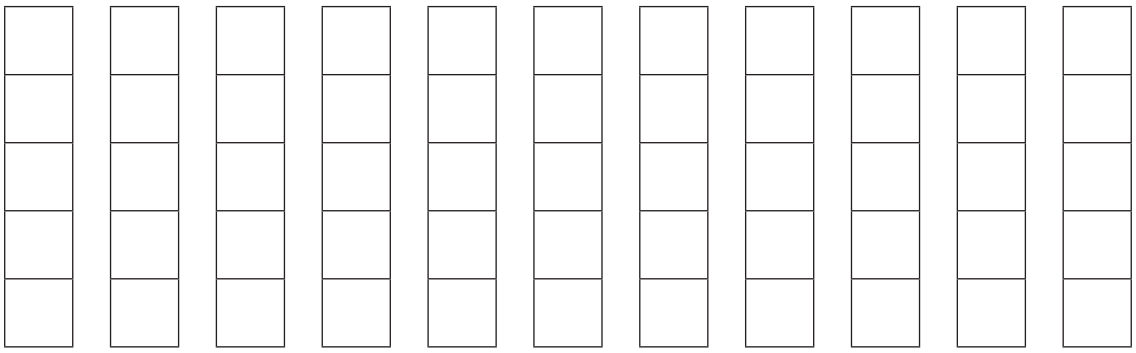
..... [2]

(c) The RPN expression

```
16  3 * 36  6 / 18  6 - * +
```

is to be evaluated.

Show the changing contents of the stack as the RPN expression is evaluated.



[4]

5 **(a)** Write this infix expression in Reverse Polish Notation (RPN).

```
16 + ((9 - 7) / (18 + 6))
```

.....

.....

.....

..... [2]

(b) Write this RPN expression in infix form.

```
6  15 + 20  3 - - 27  9 * /
```

.....

.....

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.....

.....

..... [3]

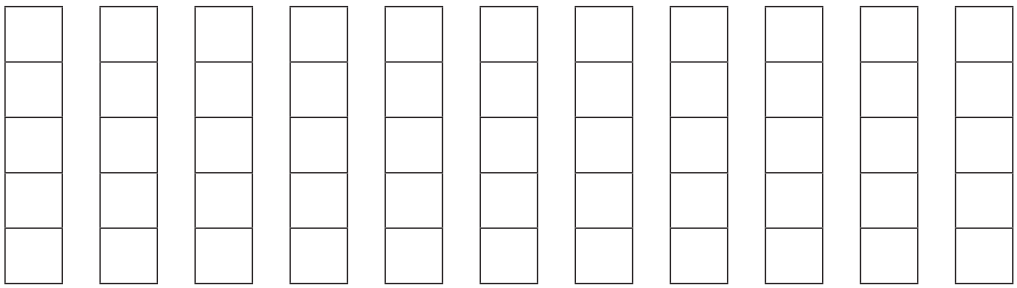
(c) The RPN expression

```
c a b - * b c + / d *
```

is to be evaluated, where:

`a = 8`, `b = 2`, `c = 10` and `d = 18`.

Show the changing contents of the stack as the RPN expression is evaluated.



[4]

6 The management and scheduling of processes are tasks carried out by an operating system.

(a) Identify **three** process states.

1 .....

2 .....

3 .....

[3]

(b) Describe the function of the shortest job first scheduling routine **and** give a benefit of this routine.

Function .....

.....

.....

.....

.....

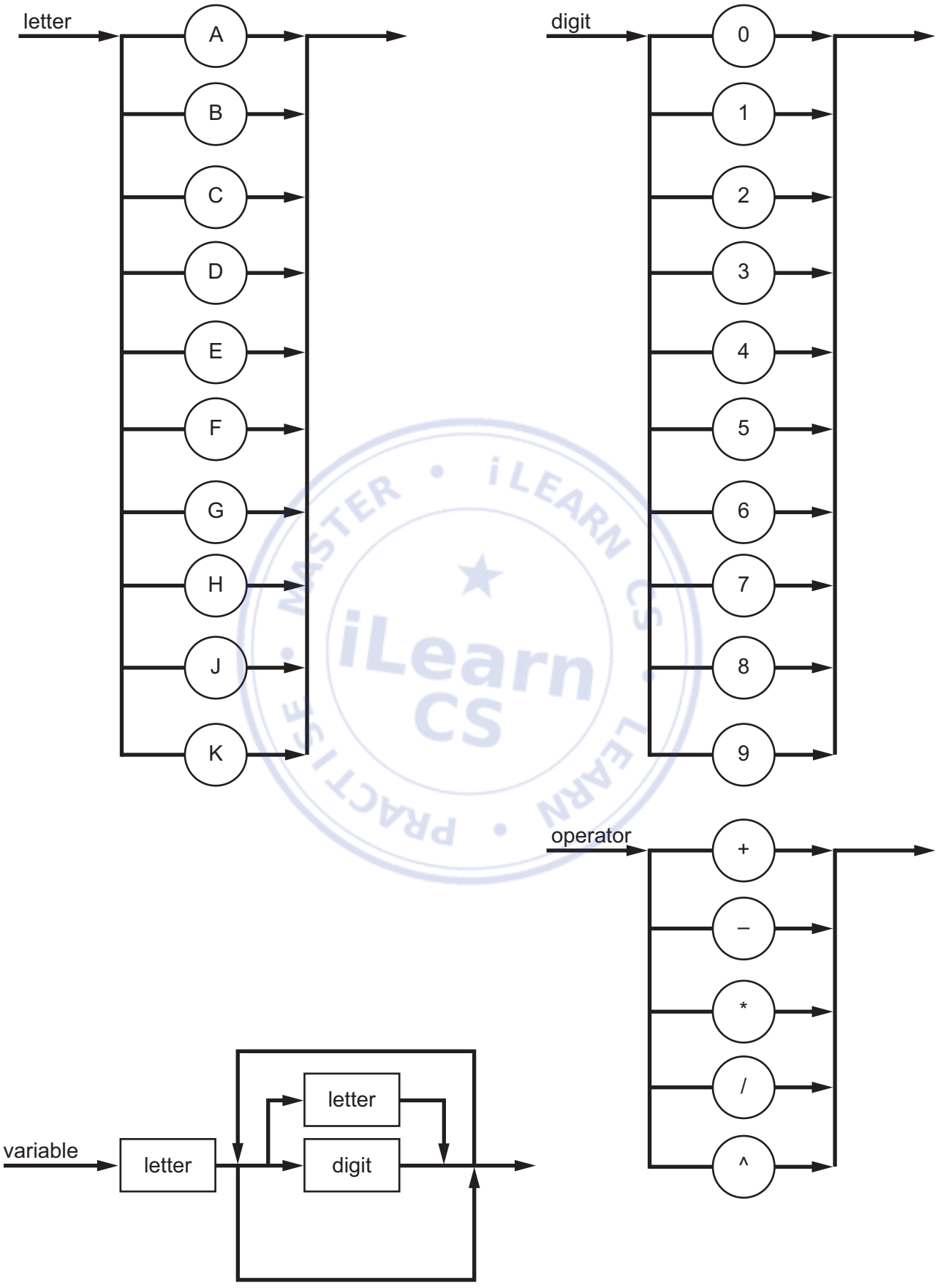
.....

Benefit .....

.....

[4]

7 Several syntax diagrams are shown.



(a) State why `9K` is **not** a valid variable for the given syntax diagrams.

.....

..... [1]

(b) Complete the Backus-Naur Form (BNF) for `<operator>`.

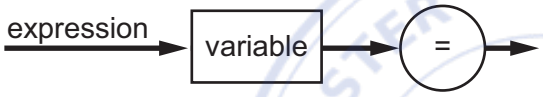
`<operator> ::=` .....

..... [1]

(c) An expression is defined as follows:

- A variable is assigned to a variable followed by an operator followed by another variable.
- The operator and final variable stage can be repeated as many times as necessary.

Complete the syntax diagram for an expression.



[3]

(d) A character can be a letter or a digit.

An additional constraint has been applied to the definition of variable. It must comply with the given syntax diagram, but it will only pass validation if it has at least **four** characters.

State **one** example of a valid variable.

.....

..... [1]

8 **(a)** Write this infix expression in Reverse Polish Notation (RPN).

```
21 * (15 - 3 + 8) / 7
```

.....

.....

.....

..... [2]

(b) Write this RPN expression in infix form.

```
c d + a b - a c - * /
```

.....

.....

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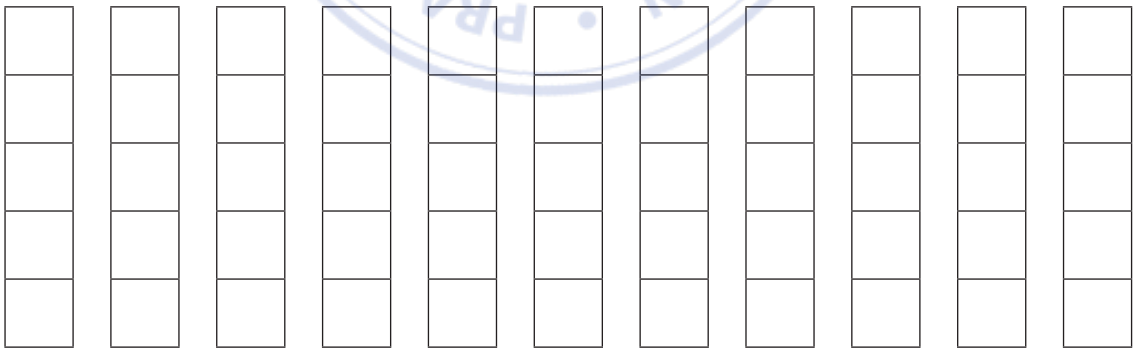
..... [3]

(c) The RPN expression

```
15  9 - 24  3 / * 18  6 + /
```

is to be evaluated.

Show the changing contents of the stack as the RPN expression is evaluated.



[4]

9 The management and scheduling of processes are tasks carried out by an operating system.

(a) Describe **one** reason why scheduling is necessary in process management.

.....

.....

.....

..... [2]

(b) Explain the function of the round robin scheduling routine **and** give a benefit of this routine.

Function .....

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Benefit .....

.....

10 Explain the process of **syntax analysis** during program compilation.

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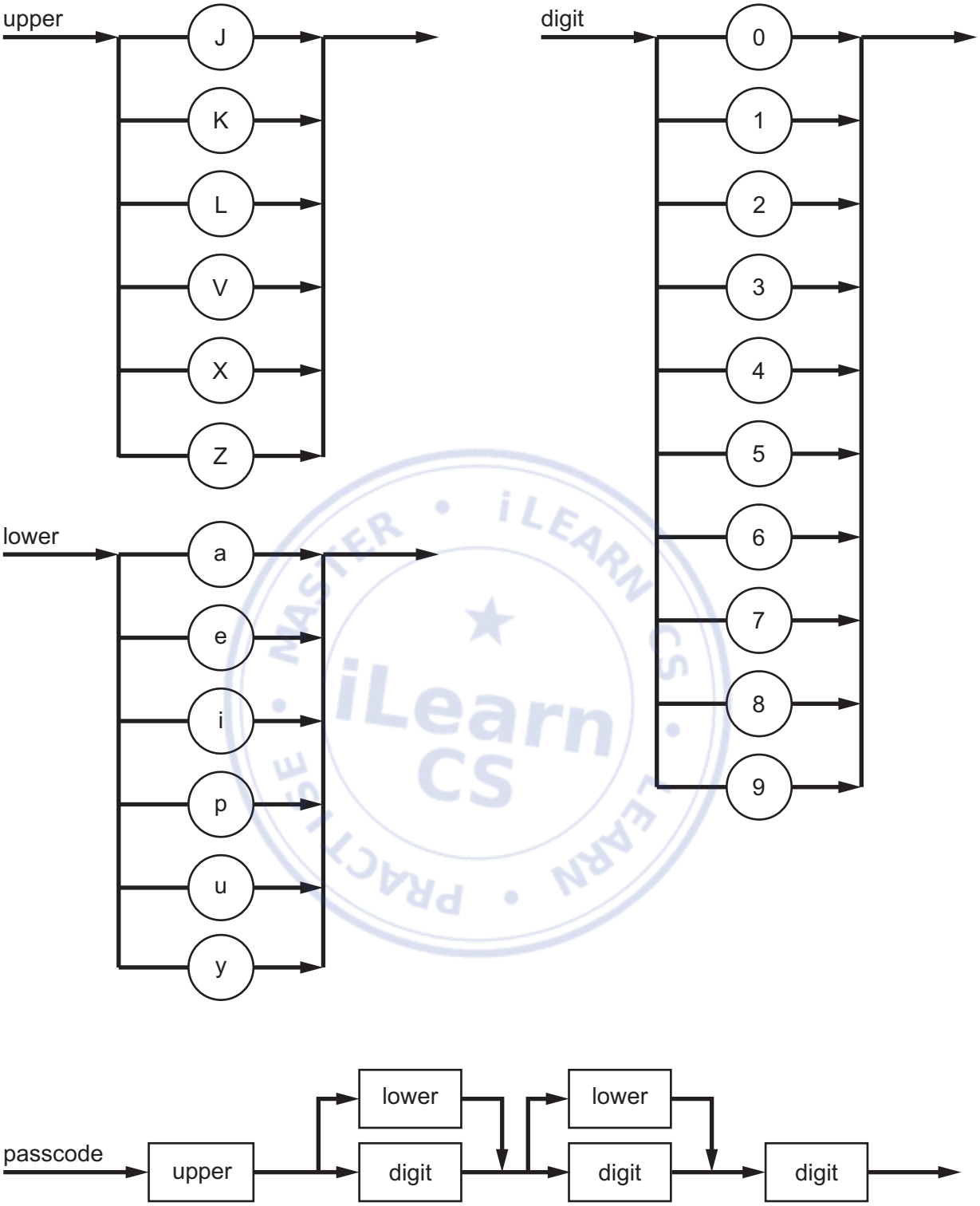
.....

.....

..... [4]

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11 Several syntax diagrams are shown.



(a) State why `JJ90` is **not** a valid passcode for the given syntax diagrams.

.....

..... [1]

(b) Complete the Backus-Naur Form (BNF) for `<upper>` and `<passcode>`.

`<upper> ::=` .....

.....

`<passcode> ::=` .....

.....

.....

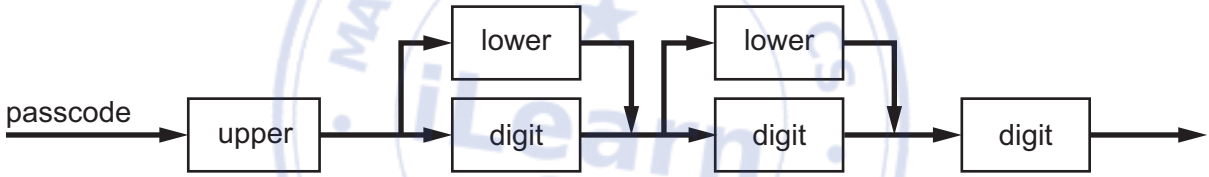
.....

[3]

(c) A character can be an upper, a lower or a digit.

The rules for passcode have been changed so that the third character may also be selected from upper and the final character may be repeated one or more times.

Complete the syntax diagram for passcode to show these changes.



[2]

END OF QUESTION PAPER

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